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Traffic Sign Classification using Deep Learning Comparative Study

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Abstract

In recent years, the advancement of deep learning technologies has significantly impacted various domains and the field of transportation is no exception. As the demand for robust and accurate traffic sign classification systems continues to rise, this study presents an in-depth exploration and comparison of various techniques employed in the field. Focusing on state-of-the-art methodologies, we assess the effectiveness and performance of multiple classification approaches for traffic sign recognition. The review presents an extensive survey of the literature, encompassing traditional computer vision methods, machine learning algorithms, and the latest advancements in deep learning. The comparative analysis aims to identify the strengths and limitations of each technique, considering factors such as computational efficiency and accuracy. Additionally, the paper implements four models—CNN, ResNet50, VGG19, and EfficientNetB7—for traffic sign classification on the GTSRB dataset, the accuracy results are reported as 99.25%, 99.28%, 98.95%, and 98.24% respectively.

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Keywords: Convolutional Neural Network, Transfer Learning, German Traffic Sign Recognition Benchmark, Traffic Sign Recognition.

1. Introduction:

Over the last few years, Intelligent Transportation Systems (ITS) have become a driving force for transformation, utilizing advanced technologies to improve the effectiveness, safety, and overall performance of transportation networks, numerous applications have been introduced under the ITS topic, such as traffic flow prediction, autonomous vehicles, accident detection and prevention, driver behavior recognition, congestion management, and traffic sign detection and classification, the last context entails the automated recognition and understanding of traffic signs, playing a substantial role in enhancing road safety and traffic management efficiency. The rapid expansion of

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road networks and the increasing complexity of traffic scenarios underscore the need for robust and efficient traffic sign recognition mechanisms. Traffic sign classification is a crucial element within intelligent transportation systems, playing a pivotal role in traffic regulation compliance of autonomous vehicles. It can also be employed for automated ticketing systems if a vehicle is detected violating traffic rules. Recognizing traffic signs related to pedestrian safety, such as crosswalks and pedestrian zones, is crucial for ensuring the safety of pedestrians and guiding drivers appropriately. It also aids emergency services in navigating through traffic efficiently. It ensures that emergency vehicles follow the correct routes and comply with traffic regulations during emergencies. Besides, it enhances road safety by improving visibility and adherence to traffic regulations that will help in reducing accidents and road safety in urban areas, urban areas, intersections, and highway areas. Moreover, it optimizes the traffic flow, by facilitating adaptive traffic control, minimizing congestion, and improving the flow of vehicles at the intersections, congested and urban areas, during peak traffic hours.

In recent years, the advent of deep learning techniques, particularly convolutional neural networks (CNNs), has revolutionized the field, offering unprecedented performance in image recognition tasks. Historically, traditional computer vision methods and rule-based systems have been employed for traffic sign detection. However, these approaches often struggle with the inherent variability in real-world conditions, such as changes in lighting, occlusions, and diverse environmental factors. The introduction of deep learning models has fundamentally transformed the landscape, as they demonstrate remarkable adaptability and generalization capabilities, crucial for addressing the complexities of traffic sign recognition. The literature surrounding traffic sign detection and classification is rich with studies that have leveraged deep learning architectures. Notable datasets, such as the German Traffic Sign Recognition Benchmark (GTSRB), and competitions like the International Joint Conference on Neural Networks (IJCNN), have provided benchmarks for evaluating the performance of these models. Researchers have explored various deep learning frameworks, including TensorFlow and PyTorch, and have developed state-of-the-art algorithms like Faster R-CNN and YOLO for real-time processing.

This paper contributes to the existing body of knowledge by synthesizing recent developments in traffic sign recognition, with a focus on the role of deep learning in general and transfer learning in particular. By critically analyzing the literature, identifying challenges, and presenting potential avenues for future research, this work aims to provide a comprehensive understanding of the current state of the art in this critical domain. It also aims to provide a comprehensive overview of the advancements in traffic sign detection and classification achieved through deep learning and machine learning. This research paper showcases an experiment on classifying traffic signs, specifically using the GTSRB dataset. Additionally, it aims to explore techniques employed by previous researchers to improve the effectiveness of the deep learning models. The goal is not only to explore these methods but also to propose architectures for developing models capable of accurately classifying traffic signs.

The subsequent sections of this paper are structured as follows. Section 2 conducts a review of pertinent literature, emphasizing pivotal studies and notable breakthroughs. Section 3 furnishes a related works comparative study on the techniques employed. Section 4 delineates the experimentation and presents the results. Ultimately, Section 5 serves as the conclusion for the paper.

2. Related works:

In this section, a spectrum of studies dedicated to these crucial components of traffic sign recognition was reviewed. The literature is categorized into three main sections: Traffic Sign Detection, Traffic Sign Classification, and Combined Approaches, each offering unique insights and contributions to the overarching goal of enhancing traffic sign recognition systems:

2.1. Traffic sign detection:

In many applications of transportation, road safety, and intelligent transportation systems, traffic sign detection is essential. The main objective of traffic sign identification is to detect and identify road signs from camera-captured photos or video streams. In this study [1] the task primarily focuses on detection. The dataset variations include TT-100K [2], MTSD [3], GTSDB, and ITSD [4], featuring public and private datasets. YOLOv3 is employed for feature extraction and YOLOv4 as the model architecture across all datasets. The reported results showcase an accuracy of 94.8% for TT-100K [2], 80.71% for MTSD [3], 91.74% for GTSDB, and 77% for ITSD [4], this study underscores

the consistent use of YOLOv3 and YOLOv4 in improving traffic sign detection across diverse unconstrained environments and datasets. In the context of object detection, a notable contribution [5] has been made in 2023 with the introduction of an enhanced YOLO (You Only Look Once) model tailored for the detection of small target images, specifically focused on traffic signs. The study utilized the TT100K [2] dataset, and incorporated the backbone of YOLOv5 for feature extraction, the proposed model, denoted as YOLOv5 SPD + Head + CAM, demonstrated commendable performance with an accuracy of 95%, precision of 91.6%, and recall of 95.4%, these results underscore the effectiveness of the model in the task of detecting small target objects within the domain of traffic sign recognition. The research [6] focuses on the task of detection using the CCTSDB dataset, this study introduces enhancements to the YOLOv5 architecture specifically tailored for the detection of small traffic signs under challenging weather conditions, the feature extraction process involves the addition of a small object detection layer, the proposed model, incorporating an Attention Mechanism with YOLOv5, demonstrated notable performance with an accuracy of 88.1%, precision of 79.8%. In [7], an extensive assessment of state-of-the-art models was conducted using the COCO [8] and GTSDB datasets, various feature extraction methods, including Inception Resnet V2, Resnet V1 101, Resnet V1 50, Inception V2, and Mobilenet V1, were coupled with models such as Faster R-CNN, R-FCN, YOLO V2, SSD Inception V2, and SSD Mobilnet, the evaluation revealed noteworthy performance, with Faster R-CNN Inception Resnet V2 achieving the highest mean Average Precision (mAP) of 95.77%, closely followed by other Faster R-CNN and R-FCN variants. YOLO V2 and SSD Mobilnet exhibited slightly lower mAP percentages, indicating the effectiveness of specific feature extraction and model combinations for accurate traffic sign detection in complex scenarios.

2.2. Traffic sign classification:

A comprehensive exploration titled [9] was conducted, focusing on the task of classification, its primary goal was to assess the impact of data augmentation on the recognition task, utilizing a custom dataset known as the Tunisian Dataset comprising 27,795 samples, the dataset is imbalanced across 49 classes, it was meticulously labeled, the study employed image augmentation techniques, including blurring, contrast adjustments, histogram modifications, and sigmoid transformations. Various advanced models were applied, including VGG16, Optimized LeNet, AlexNet, ResNet, and GoogleNet. Notably, the achieved classification accuracies were remarkable, ranging from 94.9% for VGG16, 99.6% for Optimized LeNet, 97.06% for AlexNet, 94.64% for ResNet, and 97.02% for GoogleNet, showcasing the effectiveness of the implemented techniques. A significant contribution emerged in [10], the primary task addressed in this study was classification, with the overarching goal of optimizing deep residual networks by tuning loss functions and epochs. The dataset, named Arabic Traffic Signs (ArTS) [11], comprised 2,718 publicly available samples, distributed across 24 imbalanced classes. Notably, the dataset underwent augmentation, expanding to an impressive 57,078 images. The dataset was carefully labeled, and the study employed advanced techniques such as data augmentation, enhancement, rotation, and translation. The ResNet architecture was leveraged, with variations including ResNet V1 and ResNet V2. The results demonstrated notable achievements, with ResNet V1 achieving an accuracy of 91.8%, and ResNet V2 surpassing with an impressive accuracy of 97.55%, this research underscores the effectiveness of optimized deep residual networks in addressing the complexities of Arabic traffic sign recognition, achieving high accuracy through meticulous model tuning and data augmentation strategies. In [12] The primary task focused on classification, to effectively categorize traffic signs, the study utilized the German Traffic Sign Detection Benchmark (GTSDB) dataset, it was labeled and underwent preprocessing techniques such as grayscale conversion, normalization, and histogram equalization. A convolutional neural network (CNN) architecture was employed for feature extraction and model development. The CNN model achieved a commendable accuracy of 93%, showcasing its efficacy in addressing the challenges posed by the diverse and imbalanced nature of traffic sign datasets, the newly generated dataset for testing purposes was introduced, although details about the size, public availability, balance, target count, and labeling status of this dataset remain unknown, the study [12] employed CNN architecture for both feature extraction and model development. Despite the limited information on the newly generated dataset, the CNN model achieved a recognition accuracy of 69%. A new approach [13] was presented, the primary task undertaken was classification, with the overarching goal of developing a resilient network for accurately recognizing traffic signs. The study employed two distinct datasets, namely the German Traffic Sign Recognition Benchmark (GTSRB) and the Brazilian Traffic Sign Classification (BTSC) [14] datasets. BTSC contained 7,095 publicly available samples across 62 imbalanced classes. Both datasets were labeled, and the study focused on enhancing the robustness of the

recognition system through various techniques, including data augmentation, scaling, Gaussian blur, perspective transformation, HSV adjustments, random cropping, and rotation. For feature extraction and model development, the study leveraged the ResNet architecture, demonstrating its effectiveness in handling the complexities of traffic sign recognition. Notably, the ResNet models achieved outstanding classification accuracies, with 99.51% accuracy on the GTSRB dataset and 99.22% accuracy on the BTSC dataset. These results underscore the robustness and generalization capabilities of the ResNet architecture, showcasing its suitability for embedded traffic sign recognition tasks across diverse datasets with varying class distributions. In [15] they used a custom dataset, the details of which include 2,182 samples, marked as private with unknown characteristics regarding public availability, balance, and size. The dataset featured 8 classes of traffic signs and was unlabeled. The study applied several preprocessing techniques such as resizing, normalization, labeling, and format conversion to prepare the data for deep learning analysis. For feature extraction and model development, the study employed two distinct deep learning architectures: YOLOv5 and SSD. YOLOv5 achieved an impressive classification accuracy of 97.7%, showcasing its effectiveness in recognizing traffic signs even in the absence of labeled data. Additionally, SSD demonstrated a competitive accuracy of 90.14%. These results underscore the applicability of deep learning techniques, especially YOLOv5, in traffic sign recognition tasks, even when dealing with smaller, private datasets with limited information. In [16] The primary task involved classification, to accurately categorize traffic signs. The study utilized the German Traffic Sign Detection Benchmark (GTSDB) dataset, to enhance the robustness of the classification model, the study employed various techniques, including data augmentation, grayscale conversion, histogram equalization, and normalization. These preprocessing steps were crucial for preparing the data for deep neural network analysis. The Convolutional Neural Network (CNN) architecture was chosen for feature extraction and model development. The CNN model achieved impressive performance metrics, with an accuracy of 98.44%. This result indicates the effectiveness of the proposed deep neural network approach in accurately classifying traffic signs, showcasing its potential for real-world applications, particularly in the context of road safety and intelligent transportation systems. In [17] they aimed to investigate the impact of the Adam optimizer and dropout on increasing the accuracy of the model in the task of classification. The primary goal was to explore and demonstrate the effectiveness of these optimization techniques on traffic sign recognition. The study utilized the Brazilian Traffic Sign Dataset (BTSD) [14], which included 7,095 publicly available samples across 62 imbalanced classes. The dataset was labeled and exhibited characteristics of being publicly accessible and imbalanced. The study employed Convolutional Neural Network (CNN) architecture for feature extraction and model development. Notably, the research focused on the utilization of the Adam optimizer and dropout techniques to enhance the model's accuracy. Despite the absence of specific information on precision, recall, F1 score, and mAP percentage, the CNN-based model achieved an impressive overall accuracy of 96.31%.

2.3. Combined approach:

In the pursuit of advancing the field of traffic sign detection and recognition, a contribution with a focus on the unique context of Indian traffic signs. In the paper [18] The algorithm primarily focused on the detection task and utilized the Indian traffic sign dataset. The model employed a combination of ResNet for segmentation and Convolutional Neural Networks (CNN) for classification in the feature extraction process. The model demonstrated a commendable accuracy of 98.13%. In the paper [19], the dual tasks of detection and classification were addressed with a focus on a custom dataset specifically tailored for Mexican traffic signs. A combination of R-CNN and YOLOv3 models was employed for the detection task, resulting in an accuracy of 95.33%. Additionally, the classification task utilized a ResNet model, achieving an accuracy of 90.33%. The article [20] introduces a methodology for addressing the challenges associated with detection and object tracking in real-time. The study employs a proprietary dataset containing 1,005,162 instances, encompassing 98 classes representing various traffic signs, including 36 caution signs, 30 regulation signs, 28 indication signs, and 4 other signs. Notably, the dataset is characterized by an imbalanced distribution, and despite lacking labels, the system incorporates labeling strategies. The technological components of the system involve resizing and labeling, with YOLOv5 serving as the feature extraction method. The model architecture is a fusion of YOLOv5 for detection and strongSort for object tracking. While the dataset is not publicly available and is marked as imbalanced, the system achieves commendable performance metrics. Specifically, it attains an accuracy rate of 95.8%, a precision of 97.7%, and a recall of 97%. The study [21] delves into the dual challenges of traffic sign detection and classification, employing a comprehensive

approach to the GTSD and BTSD [14] datasets. Utilizing Convolutional Neural Networks (CNN) for feature extraction, the research explores two distinct detection methods. The first method employs R-CNN, achieving a detection accuracy of 43.26%, while the second method leverages YOLOv5, yielding an improved accuracy of 59.88%. Notably, the study also focuses on traffic sign classification using a dedicated CNN model, demonstrating an accuracy of 99.2%. While the classification results indicate a robust performance, the detection methods highlight the varying efficacy of R-CNN and YOLOv5, with the latter exhibiting superior accuracy. This study [22] employed a custom dataset, specifically curated for Indian traffic sign characteristics. The feature extraction process involved the application of the Mask R-CNN with a dedicated Region of Interest (ROI) module. The proposed model, RMR-CNN (Refined Mask R-CNN), achieved a remarkable accuracy of 97.08%. In [23] they proposed a new algorithm, designed for both detection and classification tasks, harnessed the General Traffic Sign Detection Benchmark (GTSDB) dataset, which is publicly available. The feature extraction process was facilitated through Yolo. The model, simply denoted as Yolo, demonstrated exceptional performance metrics, boasting an accuracy of 95%. In the exploration of autonomous traffic sign detection and recognition. The study [24] focused on the dual tasks of detection and classification. The Saudi Arabian Traffic and Road Signs dataset from 2018 served as the foundation for this investigation. Feature extraction was carried out using a Convolutional Neural Network (CNN), and the model, also a CNN architecture, demonstrated remarkable performance with an accuracy of 96.14%. The study [25] undertook the tasks of both detection and classification using the German Traffic Sign Detection Benchmark (GTSDB) dataset. Two distinct approaches were employed for feature extraction and model architecture. In the first instance, circular traffic signs were detected using EDCircles, and recognition was accomplished using a fusion of TSD (Traffic Sign Detection) with RGB, EDCircles, and RGBNDiff, yielding an accuracy of 89%. In the second approach, Gabor filters, Local Binary Patterns (LBP), and Histogram of Oriented Gradients (HOG) were utilized for feature extraction, and the TSR (Traffic Sign Recognition) model achieved an impressive accuracy of 97.04%.

3. Related works comparative study:

Table 1 represents a summary of various research articles in the field of traffic sign detection, classification, and combined approaches. It provides information on the task addressed in each article, the datasets used, the models employed for feature extraction and detection/classification, and the reported results in terms of accuracy (acc) and, in some cases, additional metrics like precision, recall, and mAP (mean Average Precision):

Table 1: traffic sign recognition's comparative study:

Article reference	Task	Dataset	Model	Results
[1]	Detection	TT-100K	YOLOv3 for feature extraction, YOLOv4 for detection	TT-100K (acc): 94.8%
		MTSD		MTSD (acc): 80.71%
		GTSDB		GTSDB (acc): 91.74%
		ITSD		ITSD (acc): 77%
[5]	Detection	TT100K	Yolov5	Accuracy: 95%, Precision: 91.6%, Recall: 95.4%
[6]	Detection	CCTSDB	Yolov5 + attention mechanism	Accuracy: 88.1% Precision: 79.8%
[7]	Detection	COCO, GTSDB	Combination of Faster R-CNN Inception Resnet V2	Faster R-CNN Inception Resnet V2 mAP: 95.77%
[9]	Classification	Tunisian Dataset	VGG16	Accuracy: 94.9%
			Optimized LeNet	Accuracy: 96%
			AlexNet,	Accuracy: 99.6%
			ResNet	Accuracy: 94.64%
			GoogleNet	Accuracy: 97.02
[10]	Classification		ResNet V1	Accuracy: 91.8%

		Arabic Traffic Signs (ArTS)	ResNet V2	Accuracy: 97.55%
[12]	Classification	GTSDB	CNN	Accuracy: 93%.
		generated dataset		Accuracy: 69%
[13]	Classification	GTSRB	ResNet	GTSRB (acc): 99.51%
		BTSC		BTSC (acc): 99.22%
[15]	Classification	Custom Dataset	YOLOv5	YOLOv5 (acc): 97.7%
			SSD	SSD (acc): 90.14%
[16]	Classification	GTSDB	CNN	98.44% (acc)
[17]	Classification	BTSD	CNN	96.31% (acc)
[18]	Detection & Classification	Custom Indian traffic sign	Resnet for segmentation and CNN for classification	98.13% (acc)
[19]	Detection & Classification	Custom dataset for Mexican traffic signs	Detection: R-CNN + YOLOv3	Detection accuracy: 95.33%
			Classification: ResNet	Classification accuracy: 90.33%
[20]	Detection & Classification	Custom Dataset	YOLOv5 for detection, strongSort for object tracking	Accuracy: 95.8%, Precision: 97.7%, Recall: 97%
[21]	Detection & Classification	GTSD and BTSD datasets	Detection: R-CNN YOLOv5,	43.26% (acc) 59.88% (acc)
			Classification: CNN	99.2% (acc)
[22]	Detection & Classification	Indian traffic sign Dataset	RMR-CNN	97.08% (acc)
[23]	Detection & Classification	GTSDB	YOLO	95% (acc)
[24]	Detection & Classification	Saudi Arabian Traffic and Road Signs dataset	CNN	96.14% (acc)
[25]	Detection & Classification	GTSDB	Approach 1: EDCircles + TSD + RGB/EDCircles/RGBNDiff	Approach 1: 89% (acc)
			Approach 2: Gabor filters, LBP, HOG + TSR	Approach 2: 97.04% (acc)

For the detection tasks, [7] achieved the highest precision, boasting an impressive 95.77% on the GTSDB dataset through the adept combination of Faster R-CNN Inception Resnet V2. Concerning accuracy, YOLOv5 from [5] emerged as the top performer, attaining the highest accuracy of 95% on the TT100K dataset. Following closely, YOLOv4 reported the second-highest accuracy, securing a commendable 94.8% on the TT-100K dataset. In the domain of classification tasks, [13] takes the lead, with ResNet securing the top spot by achieving an outstanding accuracy of 99.51% on the GTSDB dataset and an impressive 99.22% on the BTSC dataset. Following closely, [16] features CNN as the runner-up, attaining a noteworthy accuracy of 98.44% on the GTSDB dataset. In the dual task of detection and classification, the combined model in [18], utilizing ResNet for segmentation and CNN for classification, secures the top position with an impressive accuracy of 98.13% on a custom Indian traffic sign dataset. Following closely, YOLOv5 in [20] takes the second spot for detection, achieving an accuracy of 95.8%, precision of 97.7%, and recall of 97%. In the third position, CNN from [24] demonstrates notable performance with an accuracy of 96.14% on the Saudi Arabian Traffic and Road Signs dataset (ArTS). As part of technical exploration, in the next section, we will present our implementation of the CNN and its variants using transfer learning such as VGG19, ResNet50, and EfficientNetB07

4. Methodology:

In this phase, four distinct models—Convolutional Neural Network (CNN), VGG19, ResNet50, and EfficientNetB07— were trained and evaluated on the GTSRB - German Traffic Sign Recognition Benchmark [27] dataset for the tasks of traffic sign classification. The chosen methodologies leveraged insights gleaned from the related work, guiding the customization and implementation of each model. The experiment schema is shown in fig 1

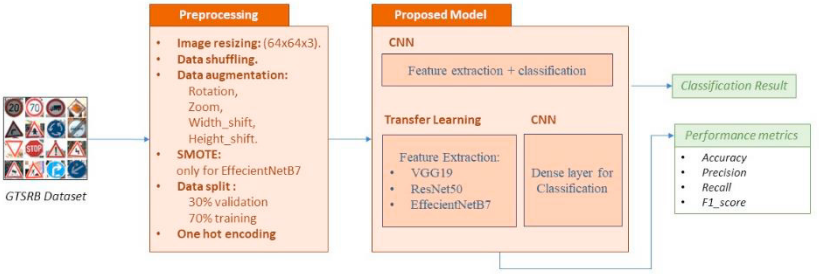


Figure 1: experiment schema.

4.1. Dataset:

Several factors contribute to the selection of the GSTDB dataset over others, with its widespread acceptance and frequent utilization for benchmarking traffic sign detection approaches in existing literature being significant considerations. Additionally, the dataset's authors, along with the supporting organization, conducted a public challenge. This initiative attracted contributions and evaluations from scientists across various disciplines, further validating the utility and reliability of the GSTDB dataset. The dataset characteristics are:

- Single-image, multi-class classification problem
- 51,839 images
- With 43 class
- The dataset is imbalanced: for example, the "Speed Limit 50 km/h" category with 2250 training images, significantly outnumbering the 200 images for the "Dangerous Curve Left" category.

4.2. Data Preprocessing:

In the initial phase of preprocessing, both the training and test datasets were imported. Subsequently, the images were resized to standard dimensions, by setting the image height and width to 64 pixels each, while also ensuring the presence of three-color channels. Following this, the shuffling to enhance the randomness of our data was implemented. Furthermore, the training data was partitioned into training and validation sets, allocating 30% to the validation set. One-hot encoding was incorporated, transforming categorical labels into a binary matrix format. Recognizing the imbalanced nature of the data, oversampling methodologies and data augmentation techniques were employed to rectify class distribution disparities, ensuring a more equitable representation of each class in the dataset, and avoiding overfitting/underfitting of certain classes. This comprehensive preprocessing strategy lays the groundwork for robust model training and evaluation in subsequent stages of this research. In the overall process, data preprocessing is a shared step that is uniformly applied across all our algorithms. This common step serves as a foundational or fundamental stage that is essential for the proper functioning of each algorithm. It establishes a baseline or a starting point that is consistent regardless of the specific characteristics or requirements of the individual algorithms.

4.3. Models:

4.3.1. CNN Architecture:

In this study, a Convolutional Neural Network (CNN) was developed for a multi-class classification task with 43 classes. The CNN architecture includes convolutional layers with filters ranging from 16 to 128, utilizing a (3, 3) kernel size and ReLU activation. Max pooling and batch normalization are applied for feature extraction and normalization. A flattening layer transforms the output for dense connected layers, with the initial dense layer comprising 512 units, ReLU activation, and batch normalization. To prevent overfitting, a dropout layer with a 0.5

rate is incorporated. The final dense layer outputs 43 classes using softmax activation. This comprehensive architecture aims to effectively address the challenges of multi-class classification, ensuring robust performance through feature extraction and model generalization.

4.3.2. VGG19:

The architecture unfolds with the incorporation of the VGG19 pre-trained model, integrating convolutional layers with varying filter sizes, ranging from 16 to 128, and employing a (3, 3) kernel size with Rectified Linear Unit (ReLU) activation. Strategic max pooling, with a window size of (2, 2), follows the second and fourth convolutional layers, bolstered by batch normalization for normalization purposes. A flattening layer precedes densely connected layers, with the initial dense layer featuring 512 units, ReLU activation, and batch normalization. To counter overfitting, a dropout layer with a rate of 0.5 is seamlessly integrated. The ultimate dense layer produces 43 class predictions with SoftMax activation, emblematic of the classification nature of the task.

4.3.3. ResNet50:

The ResNet50 architecture, a neural network pre-trained on the ImageNet dataset, was harnessed for the specific objective of traffic sign classification. The model initiated its architectural configuration with the integration of the ResNet50 base for robust feature extraction, utilizing pre-learned weights from ImageNet. Subsequent layers featured batch normalization for normalization, a flattening layer transforming output into a one-dimensional array, and a dense layer with 512 units and ReLU activation for effective feature abstraction. Following this, batch normalization was applied. To counter overfitting, a dropout layer with a rate of 0.5 was seamlessly incorporated before the ultimate dense layer, which generated 43 class predictions utilizing softmax activation for multi-class classification. To ensure the model's resilience and generalization capabilities, data augmentation techniques, including rotation, zooming, width and height shifting, and shear transformations, were implemented during training. Notably, horizontal and vertical flips were excluded to preserve the orientation of traffic signs.

4.3.4. EffecientNetB7:

In the experimental phase of the study, the EfficientNetB7 architecture was implemented, to distinguish 43 distinct classes associated with traffic signs. The learning rate was dynamically scheduled using an exponential decay strategy, initialized at 0.0001, and adapted with decay steps set at 1000 and a decay rate of 0.9 to address the problem of sticking in the local optima, fostering an adaptive optimization process. The pre-trained EfficientNetB7 model, initially trained on the ImageNet dataset, was harnessed as a feature extractor, and its parameters were fine-tuned. The model structure included batch normalization for normalization, L2 regularization for feature extraction, dropout for regularization, and a final dense layer with softmax activation for categorical predictions. To prevent overfitting, an early stopping strategy with patience of 5 epochs was implemented during training.

4.4. Hyperparameters:

Table 2 shows the hyperparameters used for each algorithm. Decay was added to prevent getting stuck in local optima and to enhance the search for global optima. The Adam optimizer has been used to optimize our models, it is an enhancement of the Stochastic Gradient Descent (SGD) algorithm, featuring dynamic adjustments of learning rates tailored to individual weights in the model.

Table 2: Hyperparameters configuration

Model	Learning rate	Decay	Epochs	Loss Function
CNN	0.001	Lr/(epochs*0.5)	40	Categorical cross-entropy
VGG19	0.001	Exponential Decay with the step of 1000 and 0.9 rate	20	Categorical cross-entropy
ResNet50	0.001	Lr/(epochs*0.5)	20	Categorical cross entropy
EfficientNetB7	0.0001	Exponential Decay with the step of 1000 and 0.9 rate	20	Categorical cross-entropy

5. Experiment and Results:

In the results section, the performance of four distinct convolutional neural network (CNN) architectures, namely the traditional CNN, VGG19, ResNet50, and EfficientNetB7, is comprehensively evaluated for the task of traffic sign classification. The CNN model represents a baseline architecture designed for feature extraction and classification. VGG19, a deeper model with a series of convolutional layers, showcases its ability to capture intricate hierarchical features. ResNet50, with its residual connections, aims to alleviate the vanishing gradient problem, potentially enhancing gradient flow during training. EfficientNetB7, known for its efficient use of model parameters, is explored for improved computational efficiency. Each model is trained and validated using meticulous hyperparameter tuning, and their performances are rigorously assessed based on metrics such as accuracy, recall, precision, and F1 score. The findings provide insights into the trade-offs between model complexity, computational efficiency, and classification accuracy, offering valuable considerations for selecting an optimal architecture in the context of traffic sign classification.

In traffic sign classification, diverse deep learning models were evaluated, and their performances across multiple subsets of the GTSRB datasets (e.g., training, validation, and testing sets) is shown in table 3.

Table 3: Performance Metrics Average.

	Results			
Dataset	Accuracy	Precision	F1_score	Recall
CNN	99.25%	99.12%	99.01%	99.06%
VGG19	98.95%	98.99%	98.80%	98.91%
ResNet50	99.25%	99.17%	99.24%	99.22%
EfficicentNetB7	98.24%	98.63%	98.73%	98.29%

In comparison to the results reported in the related work (tables 1) section on the GTSDB dataset, our Convolutional Neural Network (CNN) exhibits a noteworthy improvement with an accuracy of 99.25%, surpassing the paper's [16] reported accuracy of 98.44%. Meanwhile, our implementation of the Residual Network (ResNet50) yields an accuracy of 99.25%, aligning closely with the paper's [15] reported accuracy of 99.51%. The minimal deviation of 0.26% suggests consistency in performance between our study and the referenced literature, particularly for the ResNet model. Overall, the proposed CNN model stands out for its superior accuracy, showcasing advancements in model performance on the GTSDB dataset. When comparing the accuracies of various models (table 2) on the classification task, both the CNN and ResNet50 models achieve the highest accuracy at 99.25%. VGG19 follows closely with an accuracy of 98.95%, demonstrating strong performance but slightly lower than the top-performing models. EfficientNetB0 lags with an accuracy of 98.24%. The close alignment in accuracy between CNN and ResNet50 indicates their comparable effectiveness for the given task, while VGG19 also performs well but with a marginal difference. EfficientNetB0, while still achieving respectable accuracy, trails behind the other models in this comparison. The figures 2, 3, 4, 5 below illustrate the learning curve and loss history for the training and validation data.

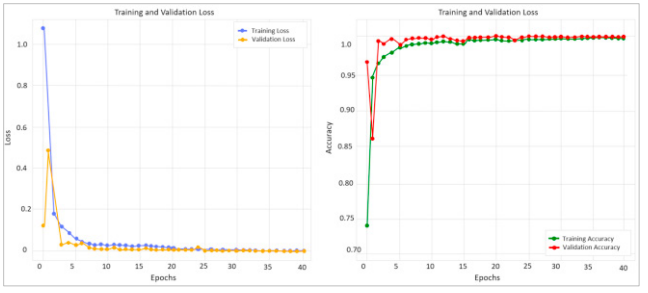


Figure 2: CNN history.

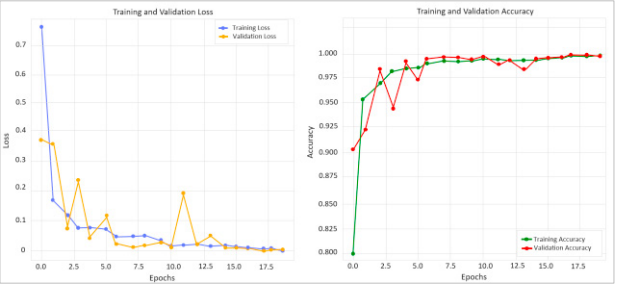


Figure 3: VGG19 learning curve and loss.

The classification error is used to assess a classification model's performance. It is sometimes referred to as a misclassification error or error rate. It shows the percentage of cases in a dataset that were erroneously classified.

Stated otherwise, it quantifies the proportion of observations, relative to the total number of observations, that the model incorrectly classified. It can be calculated by dividing Number of Misclassified Instances by total number of instances.

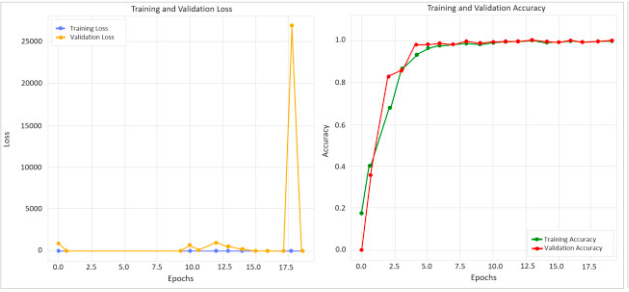


Figure 4: ResNet50 learning curve and loss.

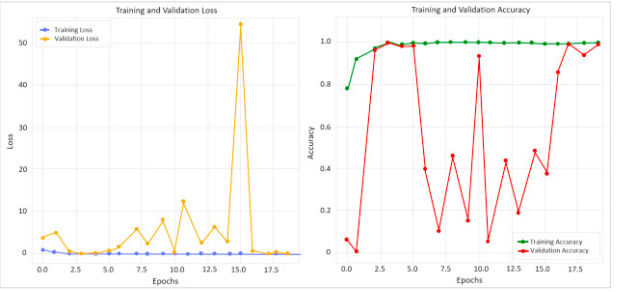


Figure 5: EfficientNetB7 learning curve and loss.

6. Conclusion:

In conclusion, a multi-faceted comparative study of state-of-the-art deep learning models across various datasets was conducted, categorizing research papers based on their focus on classification, detection, and the combined tasks of detection and classification. Subsequently, the analysis was narrowed to papers utilizing the GTSDB dataset, providing a focused examination of model performance in the context of traffic sign classification. The implementation of four prominent models—CNN, VGG19, ResNet50, and EfficientNetB0—on the classification task yielded insightful results. Notably, the proposed CNN model demonstrated exceptional accuracy at 99.25%, surpassing the comparable studies. ResNet50 also exhibited robust performance, aligning closely with the literature at 99.25%. VGG19 showcased competitive accuracy at 98.95%, while EfficientNetB0 achieved a respectable 98.24%. The nuanced differences observed across these models provide valuable insights into their relative strengths, contributing to a comprehensive understanding of model efficacy across diverse tasks and datasets. The proposed comprehensive study of deep learning models on the well-labeled and cleaned GTSRB dataset has provided valuable insights into their performance for traffic sign recognition in this paper. Moving forward, it is imperative to extend our research to more challenging datasets with real-world imperfections and variations. Additionally, future work should focus on exploring the robustness of these models under adverse conditions such as varying weather and lighting. Furthermore, while our study concentrated on classification tasks, there is an avenue for investigating these models in more complex scenarios, such as real-time detection and classification in dynamic traffic environments.

References

[1] Saxena, Swastik, Somnath Dey, Miten Shah, and Sundesh Gupta. “Traffic Sign Detection in Unconstrained Environment Using Improved YOLOv4.” Expert Systems With Applications 238 (March 1, 2024): 121836. https://doi.org/10.1016/j.eswa.2023.121836.

[2] "TT100K Dataset | Papers With Code." Accessed Nov. 30, 2023. [Online]. Available: https://paperswithcode.com/dataset/tsinghua-tencent-100k-official-training-and.

[3] "Malaysian traffic sign dataset for traffic sign detection and recognition systems | Request PDF." Accessed Nov. 30, 2023. [Online]. Available: https://www.researchgate.net/publication/313718069_Malaysian_traffic_sign_dataset_for_traffic_sign_detection_and_recognition_systems.

[4] "ISTD Dataset | Papers With Code." Accessed Nov. 30, 2023. [Online]. Available: https://paperswithcode.com/dataset/istd.

[5] Han, Tianxin, Lichao Sun, and Qian Dong. “An Improved YOLO Model for Traffic Signs Small Target Image Detection.” Applied Sciences 13, no. 15 (July 28, 2023): 8754. https://doi.org/10.3390/app13158754.

[6] Qu, Shenming, Xinyu Yang, Huafei Zhou, and Yuan Xie. “Improved YOLOv5-Based for Small Traffic Sign Detection under Complex Weather.” Scientific Reports 13, no. 1 (September 27, 2023). https://doi.org/10.1038/s41598-023-42753-3.

[7] Arcos-García, Álvaro, Juan A. Álvarez-García, and Luis Miguel Soria Morillo. “Evaluation of Deep Neural Networks for Traffic Sign Detection Systems.” Neurocomputing 316 (November 1, 2018): 332–44. https://doi.org/10.1016/j.neucom.2018.08.009.

[8] Lin, Tsung-Yi, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. “Microsoft COCO: Common Objects in Context.” In Lecture Notes in Computer Science, 740–55, 2014. https://doi.org/10.1007/978-3-319-10602-1_48.

[9] Hana Ben Fredj et al., “An Efficient Implementation of Traffic Signs Recognition System Using CNN,” Microprocessors and Microsystems 98 (April 1, 2023): 104791, https://doi.org/10.1016/j.micpro.2023.104791.

[10] Latif, Ghazanfar, Danyah A. Alghmgham, R. Maheswar, Jaafar Alghazo, Fadi N. Sibai, and Moustafa H. Aly. “Deep Learning in Transportation: Optimized Driven Deep Residual Networks for Arabic Traffic Sign Recognition.” Alexandria Engineering Journal 80 (October 1, 2023): 134–43. https://doi.org/10.1016/j.aej.2023.08.047.

[11] Latif, Ghazanfar, Jaafar Alghazo, Danyah A. Alghmgham, and Loay Alzubaidi. “ArTS: Arabic Traffic Sign Dataset.” Autonomous Traffic Sign (ATSR) Detection and Recognition Using Deep CNN 1 (March 7, 2020). https://doi.org/10.17632/4tznkn45mx.1.

[12] Bailke, Preeti A. “Traffic Sign Classification Using CNN.” International Journal for Research in Applied Science and Engineering Technology 10, no. 2 (February 28, 2022): 198–206. https://doi.org/10.22214/ijraset.2022.40224.

[13] Manzari, O. N., & Baradaran Shokouhi, S. "A Robust Network for Embedded Traffic Sign Recognition." In ICCKE 2021 - 11th International Conference on Computer Engineering and Knowledge, IEEE, 2021, pp. 447–451. doi: 10.1109/ICCKE54056.2021.9721520.

[14] Dalborgo, Vanessa Ferreira, Thiago B. Murari, Vinícius Souza Madureira, Jeremias Moraes, Vitor Magno O. S. Bezerra, Filipe Santos, Alexandre Do Nascimento Silva, and Roberto Luiz Souza Monteiro. “Traffic Sign Recognition with Deep Learning: Vegetation Occlusion Detection in Brazilian Environments.” Sensors 23, no. 13 (June 26, 2023): 5919. https://doi.org/10.3390/s23135919.

[15] Zhu, Yunyun, and Wei Qi Yan. “Traffic Sign Recognition Based on Deep Learning.” Multimedia Tools and Applications 81, no. 13 (March 7, 2022): 17779–91. https://doi.org/10.1007/s11042-022-12163-0.

[16] Vincent, M. A., Vidya, K. R., & Mathew, S. P. "Traffic Sign Classification Using Deep Neural Network." In 2020 IEEE Recent Advances in Intelligent Computational Systems, RAICS 2020, IEEE, Dec. 2020, pp. 13–17. doi: 10.1109/RAICS51191.2020.9332474.

[17] 2019 International Conference on Intelligent Computing and Control Systems (ICCS). IEEE.

[18] “Detection of Traffic Sign Using CNN.” Recent Trends in Parallel Computing, January 1, 2022. https://doi.org/10.37591/rtpc.v9i1.269.

[19] RRodríguez, Rúben Castruita, Carlos Mendoza Carlos, Osslan Osiris Vergara Villegas, Vianey Guadalupe Cruz Sánchez, and Humberto De Jesús Ochoa Domínguez. “Mexican Traffic Sign Detection and Classification Using Deep Learning.” Expert Systems With Applications 202 (September 1, 2022): 117247. https://doi.org/10.1016/j.eswa.2022.117247.

[20] Kim, Chang-Il, Jinuk Park, Jong Sun Park, Woo-Sik Jung, and Yongseok Lim. “Deep Learning-Based Real-Time Traffic Sign Recognition System for Urban Environments.” Infrastructures 8, no. 2 (January 31, 2023): 20. https://doi.org/10.3390/infrastructures8020020.

[21] Youssouf, Njayou. “Traffic Sign Classification Using CNN and Detection Using Faster-RCNN and YOLOV4.” Heliyon 8, no. 12 (December 1, 2022): e11792. https://doi.org/10.1016/j.heliyon.2022.e11792.

[22] Megalingam, Rajesh Kannan, Kondareddy Thanigundala, Sreevatsava Reddy Musani, Hemanth Nidamanuru, and Lokesh Gadde. “Indian Traffic Sign Detection and Recognition Using Deep Learning.” International Journal of Transportation Science and Technology 12, no. 3 (September 1, 2023): 683–99. https://doi.org/10.1016/j.ijtst.2022.06.002.

[23] Li, Ming, Li Zhang, Linlin Li, and Wenlong Song. “Yolo-Based Traffic Sign Recognition Algorithm.” Computational Intelligence and Neuroscience 2022 (August 3, 2022): 1–10. https://doi.org/10.1155/2022/2682921.

[24] Alghmgham, Danyah A., Ghazanfar Latif, Jaafar Alghazo, and Loay Alzubaidi. “Autonomous Traffic Sign (ATSR) Detection and Recognition Using Deep CNN.” Procedia Computer Science 163 (January 1, 2019): 266–74. https://doi.org/10.1016/j.procs.2019.12.108.

[25] Berkaya, Selcan Kaplan, Huseyin Gunduz, Özgür Özsen, Cüneyt Akınlar, and Serkan Günal. “On Circular Traffic Sign Detection and Recognition.” Expert Systems With Applications 48 (April 1, 2016): 67–75. https://doi.org/10.1016/j.eswa.2015.11.018.